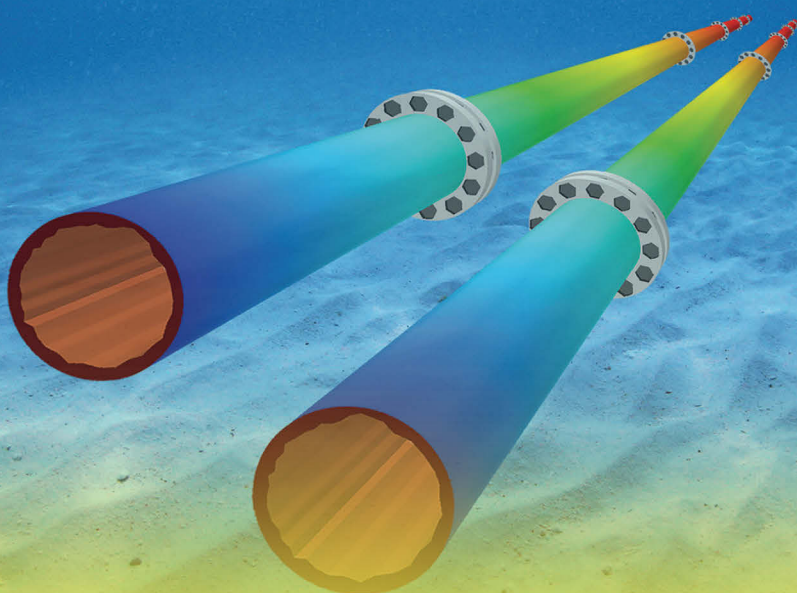


WAX DEPOSITION

Experimental Characterizations,
Theoretical Modeling, and Field Practices



Zhenyu Huang
Sheng Zheng
H. Scott Fogler



CRC Press
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Wax Deposition

Experimental Characterizations,
Theoretical Modeling, and Field Practices

Emerging Trends and Technologies in Petroleum Engineering

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Abhijit Y. Dandekar

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Contents

Series Preface	ix
Preface.....	xiii
Authors	xv
1. Introduction	1
1.1 Background of Wax Deposition	1
1.2 Overview of Wax Testing, Modeling, and Management	5
2. Experimental Characterization of Important Wax Thermodynamic Properties.....	9
2.1 Introduction	9
2.2 Determination of the WAT	10
2.2.1 Visual Inspection	12
2.2.2 CPM Techniques	14
2.2.3 Fourier-Transform Infrared Spectroscopy	17
2.2.4 Viscometry	19
2.2.5 Thermal Techniques—DSC	21
2.2.6 Comparison between Different WAT Measurements	26
2.2.7 Other Techniques That Are Under Development	29
2.3 Determination of the WPC	30
2.3.1 Differential Scanning Calorimetry	30
2.3.2 Characterization of WPC Using NMR.....	33
2.3.3 Characterization of WPC Using FT-IR.....	34
2.3.4 Separation-Based Methods for WPC Determination	35
2.4 Experimental Techniques for the Characterization of Precipitated Wax	38
2.5 Summary	40
3. Thermodynamic Modeling of Wax Precipitation	41
3.1 Introduction	41
3.2 Fundamental Thermodynamics of Wax Precipitation	42
3.3 Step 1 in Thermodynamic Modeling: Construction of the Thermodynamic Equations	44
3.4 Step 2 in Thermodynamic Modeling: Simplification of Thermodynamic Equations	49
3.4.1 Wax Thermodynamic Models Assuming Single Solid Phase.....	51
3.4.2 Wax Thermodynamic Models Considering Multiple Solid Phases by Empirical Approaches: The Conoco and Lira-Galeana Models	55

3.5	Coutinho's Thermodynamic Model—A Theoretically Comprehensive Thermodynamic Model.....	56
3.6	Industrial Practice of Wax Thermodynamic Modeling.....	58
3.6.1	Fluid Characterization: Preprocessing of Thermodynamic Modeling	58
3.6.2	Model Tuning: Postprocessing of Thermodynamic Modeling	63
3.7	Extended Applications of Wax Thermodynamic Models	64
3.8	Summary	66
4.	Wax Deposition Modeling	67
4.1	Wax Deposition Mechanisms.....	67
4.2	Molecular Diffusion as the Main Mechanism for Wax Deposition	69
4.2.1	Step 1: Precipitation of Dissolved Wax Molecules	69
4.2.2	Step 2: Formation of Radial Concentration Gradient of Dissolved Waxy Components.....	70
4.2.3	Step 3: Deposition of Waxy Components on the Surface of an Existing Deposit	71
4.2.4	Step 4: Internal Diffusion and Precipitation of Waxy Components in the Deposit	71
4.3	Overview of Wax Deposition Modeling.....	73
4.3.1	Wax Deposition Modeling Algorithm	73
4.3.2	Overview of Various Wax Deposition Models	74
4.3.2.1	Industrialized Commercial Wax Deposition Models.....	74
4.3.2.2	Academic Wax Deposition Models.....	75
4.4	Detailed Comparison of Different Wax Deposition Models	76
4.4.1	Model Dimensions.....	76
4.4.2	Hydrodynamics in the Wax Deposition Models.....	77
4.4.3	Heat Transfer Equations and Correlations.....	78
4.4.4	Mass Transfer and Deposit Growth Rate Calculations for the Wax Deposition Models	80
4.4.4.1	Mathematical Representation of the Molecular Diffusion Mechanism	80
4.4.4.2	Simplification of Wax Deposition Mechanisms	82
4.4.5	Determining the Mass Flux.....	83
4.4.5.1	Diffusion Coefficient of Wax in Oil, D_{wax}	83
4.4.5.2	Difference in the Concentration of Dissolved Waxy Components in the Bulk Oil and at the Wall, $(C_{\text{oil}} - C_{\text{wall}})$	84
4.4.5.3	Thickness of the Mass Transfer Layer, $\delta_{\text{mass transfer}}$	86
4.5	Summary	90

- 5. Introduction to Wax Deposition Experiments..... 91**
 - 5.1 Importance of Experimental Applications..... 91
 - 5.2 Wax Deposition Flow Loop 91
 - 5.2.1 Conditioning System and Pump System..... 91
 - 5.2.2 Test Section 93
 - 5.3 Deposit Characterization 94
 - 5.3.1 Measurement of the Deposit Thickness 94
 - 5.3.1.1 Pressure-Drop Technique 94
 - 5.3.1.2 Weight Measurement Technique 97
 - 5.3.1.3 Heat Transfer Technique 97
 - 5.3.1.4 Liquid Displacement–Level Detection
Technique 100
 - 5.3.1.5 Other Less Frequently Applied Techniques..... 100
 - 5.3.2 Composition Analysis of the Wax Deposit 101
 - 5.4 Cold-Finger Wax Deposition Apparatus 103
 - 5.5 Carrying Out Flow-Loop Wax Deposition Experiments 104
 - 5.5.1 Setup of Experimental Apparatus..... 104
 - 5.5.2 Oil Characterization 105
 - 5.5.3 Wax Deposition Tests 105
 - 5.5.4 Measurement of Deposit Thickness and
Characterization of Wax Deposits 105

- 6. Applying Wax Deposition Models to Flow Loop Experiments 107**
 - 6.1 Introduction 107
 - 6.2 Uncertainties in Wax Deposition Modeling..... 107
 - 6.2.1 Characterizing the Wax Precipitation Curves 108
 - 6.2.2 Empirical Correlations for Transport Phenomena..... 108
 - 6.2.3 Uncertainties in Experimental Measurements..... 109
 - 6.2.4 Appropriate Methodology for Wax Deposition
Benchmarking 109
 - 6.3 Applying Wax Deposition Models with Deposit Thickness 110
 - 6.3.1 Selecting Wax Deposition Experiments 110
 - 6.3.1.1 Experimental Facility..... 110
 - 6.3.1.2 Test Oil 110
 - 6.3.1.3 Experimental Conditions 111
 - 6.3.2 Summary of Model Performance 114
 - 6.4 Heat and Mass Transfer Analysis of the Wax Deposition
Experiments 117
 - 6.4.1 Previous Wax Deposition Experiments on the Effect
of Temperature 117
 - 6.4.2 Theoretical Analysis..... 118
 - 6.4.2.1 Dedimensionalizing the Transport Equations... 118
 - 6.4.2.2 Characteristic Mass Flux of Wax Deposition ... 120
 - 6.4.2.3 Mass Transfer Driving Force 121

6.4.3	Effect of the Operating Temperatures	122
6.4.3.1	Analysis on the North Sea Condensate	122
6.4.3.2	Analysis on Another Oil	123
6.4.3.3	Importance of the Wax Precipitation Curve.....	125
6.4.3.4	Carbon Number Distribution of the Oil	126
6.5	Applying Wax Deposition Models to Investigate Deposit Compositions	127
6.6	Summary	129
7.	Applying Wax Deposition Models for Field Predictions	131
7.1	Introduction	131
7.1.1	Wax Control Strategies for the Field	131
7.1.2	Evaluating the Severity of Wax Deposition: The Ideal vs. the Reality	133
7.2	Example 1—Single-Phase Pipe Flow	134
7.2.1	Introduction	134
7.2.2	Wax Thermodynamic Characterizations	135
7.2.3	Deposition Predictions and Pigging Frequency Design.....	136
7.3	Example 2—Gas/Oil Flow	140
7.3.1	Introduction	140
7.3.2	Wax Thermodynamic Characterization	141
7.3.3	Deposition Predictions and Pigging Frequency Design	143
7.4	Summary	145
7.4.1	Wax Thermodynamic Characterizations	145
7.4.2	Wax Deposition Modeling.....	146
7.5	Future Outlook	146
7.5.1	Improving Field Characterization Techniques.....	147
7.5.2	Collaboration between Industrial Partners.....	147
7.5.3	Develop More Rigorous Wax Deposition Modeling for Multiphase Flow Conditions.....	147
	Bibliography.....	149
	Appendix A: Nomenclature.....	159

Series Preface

This petroleum engineering book series includes works on all aspects of petroleum science and engineering but with special focus on emerging trends and technologies that pertain to the paradigm shift in the petroleum engineering field. It deals with the increased exploitation of technically challenged and atypical hydrocarbon resources that are receiving a lot of attention from today's petroleum industry, as well as the potential use of advanced nontraditional or nonconventional technologies such as nanotechnology in diverse petroleum engineering applications. These areas have assumed a position of prominence in today's petroleum engineering field. However, although scientific literature exists on these emerging areas in the form of various publications, much of it is scattered and highly specific. The purpose of this book series is to provide a centralized and comprehensive collection of reference books and textbooks covering the fundamentals but paying close attention to these emerging trends and technologies from the standpoint of the main disciplines of drilling engineering, production engineering, and reservoir engineering.

Given the dwindling supply of easy-to-produce conventional oil, rapidly climbing energy demands, the sustained ~\$100/bbl. oil price, and technological advances, the petroleum industry is increasingly in pursuit of exploration and production (E&P) of atypical or unconventional and technically challenged oil and gas resources, which may eventually become the future of the petroleum industry. Unconventional resources typically include (1) coal bed methane (CBM) gas; (2) tight gas in ultralow permeability formations; (3) shale gas and shale oil in very low permeability shales; (4) oil shales; (5) heavy and viscous oils; (6) tar sands; and (7) methane hydrates. Compared to the world's proven conventional natural gas reserves of 6600+ trillion cubic feet (TCF), the combined CBM, shale gas, tight gas, and methane hydrate resource estimates are in excess of 730,000 TCF.¹⁻³ Similarly, out of the world's total of 9 to 13 trillion bbl. of oil resources, the conventional (light and medium oil) is only 30%, whereas heavy oil, extra-heavy oil, tar sands, and bitumen combined make up the remaining 70%.⁴ In addition shale-based oil resources worldwide are estimated to be between 6 and 8 trillion bbl.⁵ As a case in point, shale-based oil production in North Dakota has increased from a mere 3000 bbl./day in 2005 to a whopping 400,000+ bbl./day in 2011.⁶ Even the most conservative technical and economic recovery estimates of the unconventional resources represent a very substantial future energy portfolio that dwarfs the conventional gas and oil reserves. However, to a large extent, these particular resources, unlike the conventional ones, do not fit the typical profile and are to some extent in the stages of infancy, thus

needing a different and unique approach from the drilling, production, and reservoir engineering perspectives.

The petroleum engineering academic and industry community is also aggressively pursuing nanotechnology with the hope of identifying innovative solutions for problems faced in the overall process of oil and gas recovery. In particular, a big spurt in this area in the last decade or so is evident from the significant activities in terms of research publications, meetings, formation of different consortia, workshops, and dedicated sessions in petroleum engineering conferences. A simple literature search for a keyword *nanotechnology* on <http://www.onepetro.org>, managed by the Society of Petroleum Engineers (SPE), returns over 250 publications dating from 2001 onward with the bulk of them in the last 5 or 6 years. Since 2008, SPE also organized three different applied technology workshops specifically focused on nanotechnology in the E&P industry. An Advanced Energy Consortium with sponsorships from some major operators and service companies was also formed in 2007 with the mission of facilitating research in “micro and nanotechnology materials and sensors having the potential to create a positive and disruptive change in the recovery of petroleum and gas from new and existing reservoirs.” Companies such as Saudi Aramco have taken the lead in taking the first strides in evaluating the potential of employing nanotechnology in the E&P industry. Their trademarked Resbots™ are designed for deployment with the injection fluids for in situ reservoir sensing (temperature, pressure, and fluid type) and intervention, eventually leading to more accurate reservoir characterization once fully developed. Following successful laboratory core flood tests, they conducted the industry’s first field trial of reservoir nanoagents.⁷

The foregoing is clearly a statement of the new wave in the petroleum engineering field, which is being created by emerging trends in unconventional resources and new technologies. The publisher and its series editor are fully aware of the rapidly evolving nature of these key areas and their long-lasting influence on the current state and future of the petroleum industry. The series is envisioned to have a very broad scope that includes but is not limited to analytical, experimental, and numerical studies and methods and field cases, thus delivering readers in both academia and industry an authoritative information source of trends and technologies that have shaped and will continue to impact the petroleum industry.

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Preface

Wax deposition has become one of the most common flow assurance problems in the petroleum industry. As petroleum resources shift from onshore reservoirs toward offshore subsea production, the industry is currently facing unprecedented challenges to maintain flow assurance for petroleum production, in which the strategy to prevent or mitigate wax deposition has become increasingly costly and complicated.

The goal to manage the issue of wax deposition involves answering the following three questions:

- Do we have a problem for this field?
- If yes, what kind of a problem is it? How bad is it?
- How can we solve this problem?

They are typical questions to be answered not only for wax-related issues but also for a variety of many other common production chemistry problems in general flow assurance practices. The answer to the first question generally involves only a few fluid testing procedures that are relatively simple, while much more understandings on production chemistry and fluid flow are required to address the second question. The answer to the third question requires not only knowledge that are shown in chemical engineering textbooks but also significant operational experience, and the decision makers have to fully understand the production capability of the field and the effectiveness as well as implication of any mitigation/remediation methods.

While there are several books that provide general knowledge on flow assurance, a book that specifically addresses the issue of wax deposition is still not yet available. This book is the first one that covers the entire spectrum of knowledge on wax deposition phenomena. It provides a detailed description of the thermodynamic and transport theories for wax deposition modeling and a comprehensive review of the laboratory testing to help establish appropriate control strategies for the field. It provides a progressive introduction to help flow assurance engineers to understand the process of wax deposition, to be familiar with the various methods to identify its severity, and to eventually control this problem. For engineering students, practicing engineers, and researchers in the field of flow assurance, this book serves as an in-depth discussion of how fundamental principles of thermodynamics, heat, and mass transfer can be applied to solve a problem common to the petroleum industry.

Going back to the three key questions that were raised earlier in this Preface, we hope to provide valuable information in this book that could help the readers to address these questions. Chapter 1 presents the background

of wax deposition, including the cause of the phenomena, the magnitude of wax deposition problems, as well as its impact on petroleum production. Chapters 2 and 3 introduce various laboratory techniques and theoretical models. These testing and modeling are indispensable to address the first question (Do we have a wax problem?). Chapters 4–6 present the knowledge that is critical to answer the second key question (How bad is the problem?). In Chapter 4, a systematic presentation will be made to describe the process of wax deposition using chemical engineering fundamentals. It discusses various models of wax deposition and analyzes the differences between the assumptions used in these models. In addition, the advantages and disadvantages in each model are compared. Chapter 5 provides a detailed description of how to conduct laboratory wax deposition experiments in order to benchmark different wax deposition models. In this chapter, the applications of the cold finger apparatus and the lab scale flow loop are highlighted. Chapter 6 discusses examples of how fundamental principles of heat and mass transfer can be applied to interpret laboratory wax deposition experiments to better understand wax deposition behaviors and eventually predict the wax deposit growth in field operations. Chapter 7 brings the readers to the “real world” by providing several field examples of how management strategies for wax deposition in the field can be established based on the available laboratory testing and modeling work, thereby addressing the third question (How can we solve the problem?).

This book contains comprehensive knowledge of wax deposition not only from academic research but also from the flow assurance industry, thanks to the comments and suggestions from many petroleum companies in the industry. We acknowledge Tommy Golczynski and Tony Spratt from Assured Flow Solutions LLC for carefully reviewing the drafts of the book. We thank all the sponsors of the University of Michigan Industrial Affiliates Program, including Chevron, ConocoPhillips, Multichem, Nalco, Shell, Statoil, Total, and Wood Group Kenny. In addition to their financial support to the academic research to the Michigan Industrial Affiliates Research Program, the expertise and experience shared by the representatives of these companies constitute an integral part of the completion of this book. We also thank all the members of Professor Fogler’s research group for their effort dedicated to the Michigan Industrial Affiliates Research Program. Finally, we extend our gratitude to our family members for their support in completing this book.

Authors



Dr. Zhenyu Huang (Jason) is currently a senior flow assurance specialist in Assured Flow Solutions LLC, providing engineering solutions for a variety of flow assurance issues to the petroleum industry. His expertise includes production chemistry and multiphase flows. Dr. Huang has more than 8 years of academic and industrial experience focused extensively on a variety of wax deposition problems. His work includes model development, experimental verifications, fluid testing and field applications. He has been involved with

multiple offshore developments that present wax deposition/gelation concerns. He is the subject matter expert on wax-related issues, and he currently serves as the vice president of the Upstream Engineering and Flow Assurance Section of the American Institute of Chemical Engineers. Dr. Huang earned his bachelor's degree in Tsinghua University in Beijing, China, in 2006, and he completed his PhD study at the University of Michigan, Ann Arbor in 2011 with a thesis entitled "Application of the fundamentals of heat and mass transfer to the investigation of wax deposition in subsea pipelines."



Sheng Zheng (Mark) graduated *summa cum laude* from the University of Michigan with a bachelor's degree in chemical engineering and minors in chemistry and mathematics. He is currently a doctoral candidate in Professor Fogler's research group, specializing in both cutting-edge experimental characterizations and theoretical modeling for wax deposition research. He has multiple high-quality publications focusing on compositional wax deposition modeling and wax transport in multiphase flow conditions. Together with Dr. Huang

during their work at Wood Group Kenny, Mark carried out the DeepStar Wax Prediction and Pigging Design project, a joint industrial project to comprehensively survey and assess current industrial wax management and control strategies.



Dr. H. Scott Fogler is the Ame and Catherine Vennema professor of chemical engineering and the Arthur F. Thurnau professor at the University of Michigan in Ann Arbor; he was the 2009 president of the American Institute of Chemical Engineers. He earned his BS degree from the University of Illinois and his MS and PhD degrees from the University of Colorado. He is also the author of the *Elements of Chemical Reaction Engineering*, which is one of the main textbooks for chemical engineering students.

Scott and his students are well known for their work on the application of chemical reaction engineering principles to the petroleum industry. They have published over 200 research articles in areas such as wax deposition/gelation kinetics in subsea pipelines, asphaltene flocculation/deposition kinetics, scale deposition and acidization of petroleum wells. In 1996, he was a recipient of the Warren K. Lewis award from the American Institute of Chemical Engineers for his contributions to chemical engineering education. He is also a recipient of 11 named lectureships.

1

Introduction

1.1 Background of Wax Deposition

Wax deposition is a critical operational challenge to the oil and gas industry. As early as 1928, wax deposition was reported as an issue that “presents many difficult problems while being produced, transported, and stored” (Reistle, 1932, page 7). Wax deposition problems occur in a wide range of locations in the petroleum production chain, including flow lines, surface equipment, and topside facilities, and downstream refineries. In some of the extreme cases, it can also occur in well tubings.

The waxy components of crude oils, also known as n-paraffins, represent a group of n-alkanes with carbon numbers that are usually greater than 20 (Lee, 2008). These components are normally dissolved in the oil at reservoir conditions where the temperature is relatively high. However, as the crude oil leaves the reservoir and travels toward processing facilities, its temperature can decrease substantially and potentially fall below the wax appearance temperature (WAT) (Berne-Allen & Work, 1938). When the waxy components can precipitate out of the oil and form solids, resulting in slurries in the oil flow that require higher pressure drop for transportation. More importantly, the precipitation of these components on the inner surface of the pipe wall can lead to the formation of wax deposits, which often occurs on the tubing, the pipelines, and the process equipment (Reistle, 1932).

In early- to mid-1990s, the problem of wax deposition usually occurred during petroleum production on land or onshore resources (Reistle, 1932). In 1969, it was reported that the cost for wax control in U.S. domestic production amounted annually to \$4.5–\$5 million (Bilderback & McDougall, 1969). Because of easy access and management for these resources, the problem of onshore wax deposition can be addressed by relatively simple methods, including the optimization of the operating conditions (pipeline size, pressure, etc.). Heating of the pipeline or mechanical removal of the wax deposit was used occasionally and was generally not as prohibitive. It is during the late twentieth century that the problem of wax deposition has become increasingly challenging, as the production of petroleum fluids shifted from onshore resources toward offshore reservoirs around